

Unit 8: Quadratic Functions

1. Rewrite $y = x^2 + 22x - 53$ in vertex form.

$$\begin{array}{rcl}
 x^2 + 22x - 53 & = & 0 \\
 +53 & +53 & \\
 \hline
 x^2 + 22x & = & 53 \\
 \div 2 \downarrow & \nearrow + & \\
 (11)^2 & & \\
 \hline
 x^2 + 22x & &
 \end{array}
 \qquad
 \begin{array}{rcl}
 (x+11)^2 & - & 174 \\
 \hline
 \end{array}$$

2. Rewrite $y = -1(x+2)^2 + 8$ in standard form.

$$\begin{array}{rcl}
 (x+2)(x+2) & & \\
 x^2 + 2x + 2x + 4 & & \\
 -1(x^2 + 4x + 4) + 8 & & \\
 -x^2 - 4x - 4 + 8 & & \\
 \hline
 -x^2 - 4x + 4 & &
 \end{array}$$

3. Label the key features on the graph.

